

Benchmarking the Jacobi-Prior / Jacobian Closed-Form Estimator on the PowerCo Churn Dataset

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1. Objective

To test the practical behaviour of the closed-form Binomial-logit estimator described in Das & Dey (2006) and extended in the Jacobi Prior framework (Das & Sardar, 2024), by benchmarking it against a Random Forest baseline on a real, previously-completed classification project: SME customer churn prediction for an energy provider (PowerCo dataset, BCG X Data Science Simulation).

2. Method

- **Data:** 14,606 customers, 61 predictive features after the original feature-engineering pipeline (excluding id columns), 9.7% churn rate.
- **Split:** identical 75/25 stratified train/test split (random_state=42) used by the original Random Forest notebook, for a like-for-like comparison.
- **Estimator implemented:** the flat-prior Binomial-logit closed form from Das & Dey (2006), Table 1: $\beta = (X'X)^{-1} X' \log((y+1)/(2-y))$, applied to the Bernoulli churn indicator.
- **Baseline:** the original Random Forest configuration (1000 trees, class_weight='balanced', max_features='sqrt'), re-fit in the same environment for a fair runtime comparison.

3. Results

Metric	Jacobian / Jacobi-Prior (closed-form)	Random Forest (baseline)
ROC-AUC	0.650	0.707
PR-AUC	0.175	0.305
Precision / Recall / F1 (top-9.7% risk threshold)	0.231 / 0.231 / 0.231	0.327 / 0.327 / 0.327
Fit time	0.009 sec	21.52 sec
Inference time	0.0001 sec	0.47 sec

4. Discussion

- **Speed:** the closed-form estimator fits ~2,300x faster and infers ~4,700x faster than the Random Forest baseline, strongly reproducing the central computational-efficiency claim of the Jacobi Prior paper.
- **Accuracy:** the Random Forest baseline outperforms the closed-form estimator on every predictive metric tested. The gap is not small (ROC-AUC 0.650 vs 0.707; PR-AUC 0.175 vs 0.305). This is expected: the estimator is fundamentally a linear model in the (transformed) log-odds, while this dataset has 61 engineered, correlated, and likely nonlinear features that a 1000-tree ensemble can exploit more effectively.
- **Class imbalance:** at the default 0.5 probability threshold, the closed-form estimator assigns no customer a predicted probability above 0.5, since it has no built-in class-imbalance correction (unlike Random Forest's

class_weight='balanced'). Reported Precision/Recall/F1 use a fairer rank-based threshold instead (flagging the riskiest 9.7% of test customers, matching the true churn rate).

5. Deviations from the Original Formula (disclosed)

- **Standardization:** features were standardized (zero mean, unit variance) before applying the estimator. This is not part of the original closed-form derivation, but was necessary here: the 61-feature engineered set is collinear enough that the raw $(X'X)$ matrix is close to singular.
- **Pseudoinverse:** due to that near-singularity, the Moore-Penrose pseudoinverse was used in place of a literal matrix inverse. This is the standard, minimum-norm generalisation and is exact when $(X'X)$ is full rank, but is worth flagging as a departure from the literal formula.

6. Conclusion

On this real, moderately high-dimensional, engineered-feature dataset, the closed-form Jacobian/Jacobi-Prior estimator delivers a substantial and clearly reproducible speed advantage, but a real and non-trivial accuracy gap relative to Random Forest. This is a useful, honest data point: it suggests the estimator's competitive-accuracy results in the original paper (SDSS classification, spinal MRI features) may depend on the feature space being more linearly separable than this churn dataset's collinear, engineered feature set — a possible direction for further investigation (e.g. combining the closed-form estimator with a nonlinear basis expansion, or testing on the raw un-engineered features instead).